

Fake review detection using transformer-based enhanced LSTM and RoBERTa

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ABSTRACT

Internet reviews significantly influence consumer purchase decisions across all types of goods and services. However, fake reviews can mislead both customers and businesses. Many machine learning (ML) techniques have been proposed to detect fake reviews, but they often suffer from poor accuracy due to their focus on linguistic features rather than semantic content. This paper presents a novel semantic- and linguistic-aware model for fake review detection that improves accuracy by leveraging advanced transformer architecture. Our model integrates RoBERTa with an LSTM layer, enabling it to capture intricate patterns within fake reviews. Unlike previous methods, our approach enhances the robustness of fake review detection and authentic behavior profiling. Experimental results on semi-real benchmark datasets show that our model significantly outperforms state-of-the-art methods, achieving 96.03 % accuracy on the OpSpam dataset and 93.15 % on the Deception dataset. To further enhance transparency and credibility, we utilize Shapley Additive Explanations (SHAP) and attention techniques to clarify our model's classifications. The empirical findings indicate that our proposed model can offer rational explanations for classifying specific reviews as fake.

1. Introduction

In recent years, the Internet has greatly impacted people's lives. Consumers often spend considerable time browsing for product information, reading reviews, and interacting with others. In addition, the Internet allows users to write their evaluation remarks or reviews on various subjects according to their experience with some products or services. Those users can provide their opinions and reviews to criticize or promote a variety of goods and services (Saumya, Singh, Baabdullah, Rana, & Dwivedi, 2018; Singh, et al., 2017). Customers usually purchase goods with several positive reviews, which can increase the vendor's income (Saini, Saumya, & Singh, 2017). Moreover, unfavorable ratings can result in financial damages for the involved companies (Ho-Dac, Carson, & Moore, 2013; Hakak, et al., 2021; Khan, et al., 2021). Since anybody can submit reviews, unfair positive or negative comments about items, services, and businesses are possible (Mohawesh, Liu, Arini, Wu, & Yin, 2023; Maqsood, Xu, Springer, & Mohawesh, 2021;

Mohawesh, Tran, Ollington, & Xu, 2020; Mohawesh, Xu, Springer, Al-Hawawreh, & Maqsood, 2021; Mohawesh, Al-Hawawreh, Maqsood, & Alqudah, 2023; Mohawesh, Maqsood, Jararweh, & Salameh, 2023; Mohawesh, Maqsood, & Althebyan, 2023; Myers, et al., 2023). To help individuals from being misled by incorrect information, internet content and reviews must be verified.

There are two main methods to verify internet options and reviews: Human and automated detection. The work in (Ott, Choi, Cardie, & Hancock, 2011) stated that, around 57 % of the time, human judges could not detect fake reviews. Therefore, manual identification is not practical. It has been established that automated detection using intelligent technology can be quicker and more accurate than human experts. Several research efforts have been conducted to create automatic fake review detection models using traditional ML techniques, some of which have performed relatively well.

Nevertheless, most of these models have shortcomings, resulting in poor precision in distinguishing between fake and authentic reviews.

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Specifically, to detect fake reviews, existing models were designed based on language characteristics and have failed to capture the reviews' semantic content. Hence, new semantic-based methods that effectively detect fake reviews using ML are still necessary. Lately, transformers or sophisticated pre-trained architectures have garnered significant interest in tasks related to text classification and achieved improved outcomes compared to the existing state-of-the-art methods (Lan, et al., 2019; Liu, et al., 2019; Yang, et al., 2019). A novel semantic-based model for detecting fake reviews can be developed with the help of transformer architectures or methodologies due to their effectiveness in building deep contextualized embeddings for different types of texts. Nevertheless, the best match for the complete training data could not be achieved based on a single model due to the variability and dynamism of fake review variation attributes (Mohawesh, Liu, et al., 2023; Mohawesh, et al., 2020; Mohawesh, Xu, Springer, et al., 2021; Mohawesh, Xu, et al., 2023; Mohawesh, Xu, Tran, et al., 2021; W. Khan & Haroon, 2022). An individual transformer could be too very sensitive or biased towards some input features, leading to sub-optimal performance.

In summary, the key contributions of this paper can be summarised as follows:

1. Introducing a hybrid strategy that employs the RoBERTa transformer and LSTM architectures. The proposed approach categorizes reviews as either genuine or fake according to their syntactical, grammatical, and semantic characteristics.
2. Integrating the RoBERTa model with an LSTM to identify real or fake reviews. The proposed technique utilizes accuracy, precision, recall, and F1 score as assessment criteria to determine the leniency of the proposed model.
3. Conducting extensive experimental results to investigate the performance of the proposed model compared to several state-of-the-art methods. The results reveal that superior performance is obtained by the proposed model in terms of accuracy compared with the state-of-the-art approaches. The proposed model is shown to be more accurate than ten recent models developed to identify fake reviews.

The rest of the paper is organized as follows: Section 2 provides the related work. Section 3 elaborates on the proposed methodology. Section 4 describes the experimental setting, datasets, pre-processing, and evaluation metrics. Section 5 presents the results and analysis. Section 6 concludes the paper with the main findings.

2. Related works

Several previous studies have attempted to identify fake reviews by employing classic ML algorithms such as Naive Bayes (NB) and Support Vector Machine (SVM), which can learn discriminant features from reviews (Cardoso, Silva, & Almeida, 2018). A good instance of this is when someone employs a linguistic inquiry, and word count (LIWC) instrument (Krishnamoorthy, Sathiyarayanan, & Proença, 2024) (Pennebaker, Boyd, Jordan, & Blackburn, 2015). Ott, et al. (2011) suggested a methodology to automatically detect fraudulent reviews by combining psychological factors extracted from reviews with n-grams and feeding the resulting data to support vector machines. The method showed an accuracy of 90 % in classifying false reviews, which is significantly higher than human evaluators. Human judges achieved 57 % accuracy using the Op-Spam dataset. The proposed approach in Feng, Banerjee, and Choi (2012) detected fake reviews using part-of-speech characteristics and context-free grammar. The reported findings demonstrated that using these features in combination with the existing system considerably improved the accuracy of identifying fake reviews. Later, Li, Ott, Cardie, and Hovy (2014) proposed a model called Sparse Additive Generative Model (SAGE) to detect false reviews that employ topic modeling (DePaulo, et al., 2003) with a generalized additive model (Hastie Trevor and Tibshirani Robert 1990). The accuracy of the proposed model on the deception and Op-Spam datasets was 83.10 % and

81.82 %, respectively.

Cagnina and Rosso (2015) used character n-grams, emotion traits, and LIWC to detect fake reviews. Subsequently, an SVM algorithm is fed these extracted features to classify the reviews. An aspect of deep language that is based on a model for identifying bogus reviews was proposed by Xu and Zhao (2012). Next, they classified reviews using an SVM classifier. Fusilier, Montes-y-Gómez, Rosso, and Cabrera (2015) suggested a model for detecting fake reviews based on character n-gram content attributes. Then, NB was utilized to detect fraudulent reviews. Recently, a wide variety of deep learning models, including convolutional neural networks (CNN), recurrent neural networks (RNN), and gated neural networks (GRU), have been widely used in different applications and have produced favorable outcomes (Mohawesh, Maqsood, Jararweh, et al., 2023; Rifai, et al., 2023; Yin, Liu, Wu, Arini, & Mohawesh, 2023). Extraction of the semantic presentation from dimensional data was the focus of these models. Ren and Zhang (2016) presented a recurrent convolutional neural network approach with an attention mechanism to learning document representation. As expected, the suggested methodology successfully identified fake reviews. Moreover, Yafeng Ren and Donghong Ji (2017) proposed a hybrid fake review detection algorithm using deep neural networks and incorporating phrase context information (GRNN–CNN). They used both CNN and GRU. A successful test of their suggested model on the deception dataset yielded 83.34 % accuracy. Afterward, Zhang, Du, Yoshida, and Wang (2018) created a model for recurrent convolutional deep neural networks (DRI-RCNN) to detect false reviews by analyzing the circumstances in which words appear. The DRI-RCNN has demonstrated effective performance of accuracy levels of 80.8 %, and 82.9 % on the deception and spam datasets, respectively. The authors (Mohawesh, Xu, Tran, et al., 2021) examined several promising ML approaches to detect fake reviews. Their experimental findings showed that the RoBERTa transformer outperformed state-of-the-art approaches with a deception dataset accuracy of 91.02 percent. In addition, they illustrated that RoBERTa, DistilBERT, and BERT achieved better results with small datasets. Although several ML algorithms have been designed to detect false reviews and distinguish between real and fake reviews, contextualized text representation models were not well investigated. This paper provides a novel model that could be utilized with any contextualized text representation and neural classifier.

3. Methodology

The proposed model incorporates pre-processing and blends state-of-the-art transformer designs with an LSTM strategy as shown in Fig. 1. The following subsequent sections elaborate on these phases and the structure of the proposed model:

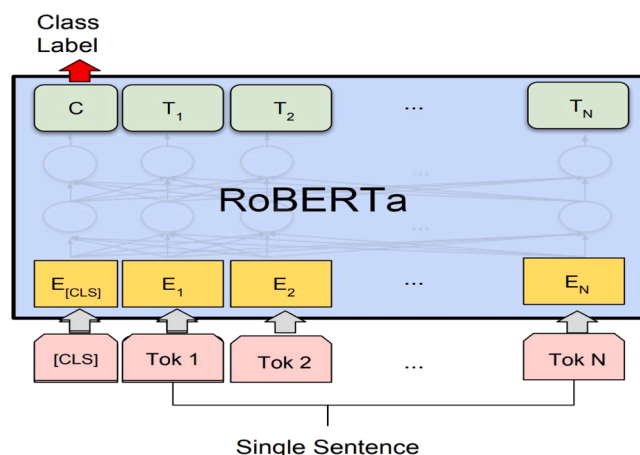


Fig. 1. The architecture of the RoBERTa model.

3.1. Pre-processing stage

Several procedures are carried out to clean the incoming review data and prepare it for use as input for the transformers. We filter unnecessary information and terms like emojis and URLs throughout this process. Finally, we consider each piece of feedback and separate it into individual words.

3.2. Robustly optimised BERT approach (RoBERTa)

RoBERTa is an improved BERT that is more powerful than the original BERT (Alippi et al., 2011). It trains the machine on longer sequences and skips sentence prediction. RoBERTa is a pre-trained model consisting of almost 63 million news items in English, English Wikipedia, and book corpora. It is employed because of its high-quality masking patterns and flexibility in adapting to each review.

3.3. Long short-term memory (LSTM)

LSTM is a type of RNN architecture and is currently a mainstream RNN structure (Hochreiter & Schmidhuber, 1997). LSTM solves the vanishing gradient by exchanging memory blocks with hidden self-connected units. The memory block is used to store information and learn long-range text sequences. The memory unit tells the network what to learn and what to forget. LSTM consists of four components (the input gate, the forget gate, the output gate, and the cell activation vector). First, the input gate manages the size of the extra memory content. Second, the forget gate f , w specifies a certain amount of the memory that must be erased. Third, the output gate o , w modulates the memory content output amount. Fourth, the cell activation vector c , contains two components, modulated new memory $\tilde{c}_t$ and forgotten previous memory c_{t-1} . t identifies the t^{th} time. With 100-dimensional GloVe embeddings, our LSTM model was trained. The dimensions and time steps of the output were both set to 300. An ADAM optimizer with a learning rate of 0.001 was used to reduce binary cross-entropy loss. The activation function for the final output layer was sigmoid. The model was trained across ten epochs with 64 and 512-element batches. Several runs were repeated with different numbers of layers for the LSTM approach.

The LSTM architecture is given in Fig. 2, where the mathematical equations of LSTM operations are shown as follows:

$$i_t = \sigma(W_{xi}x_t + W_{hi}h_{t-1} + b_i)$$

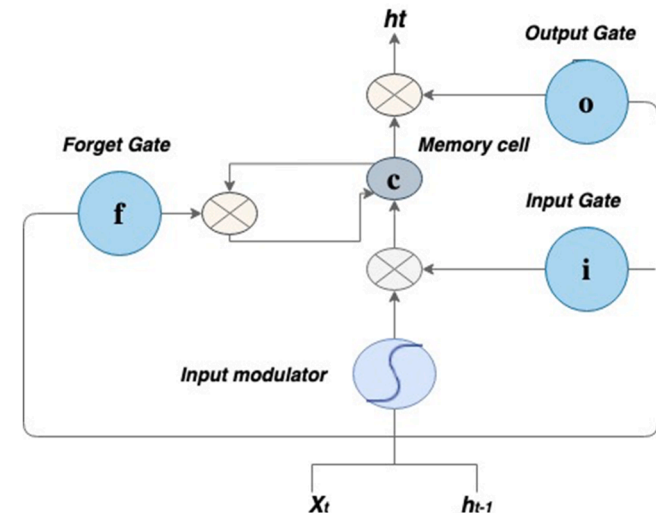


Fig. 2. The architecture of LSTM. Arrowed lines describe the weight matrices.

$$f_t = \sigma(W_{xf}x_t + W_{hf}h_{t-1} + b_f)$$

$$o_t = \sigma(W_{xo}x_t + W_{ho}h_{t-1} + b_o)$$

$$\tilde{c}_t = \tanh(W_{xc}x_t + W_{hc}h_{t-1} + b_c)$$

$$c_t = f_t \otimes c_{t-1} + i_t \otimes \tilde{c}_t$$

$$h_t = o_t \tanh \otimes (c_t)$$

where f_t , i_t , c_t , o_t reflect the values of f , i , c , o at the time t , respectively. The term W describes the hidden layer updating weights and b represents the bias. The $\tanh$ and σ represent the hyperbolic tangent and sigmoid functions, respectively.

3.4. The proposed model

The proposed model employs a feed-forward network with 768 hidden sizes trained with the RoBERTa- base-uncased model. The input IDs and attention masks of the sentence being modeled are used in the RoBERTa model. The RoBERTa tokenizer is employed for this purpose; it accepts the input sequence as [CLS] and [SEP] appended to the beginning and end of a sentence, respectively. Then, it returns the input IDs and attention masks for each sentence. These data are then used as input to the RoBERTa model, which generates an embedding vector for each token of a hidden size of 768. To improve LSTM's grasp of sentence semantics, RoBERTa offers contextualized representations at the sentence level. In recent works, researchers have demonstrated how combining LSTM with word embedding models can significantly improve detection performance (Goyal, Du, Ott, Anantharaman, & Conneau, 2021). Since combining LSTM with RoBERTa can improve prediction accuracy, it follows that the suggested model has a deeper grasp of semantic meaning. As a "collective representation" for classification tasks, [CLS] is encoded via a multi-layer encoding procedure that incorporates all representative information of all tokens. This means that the [CLS] token's embeddings can be used to represent the entire sentence and then fed into a classifier to perform a classification task. The classifier is coded from the ground up. The review text has been classified using a classification model composed of a RoBERTa and an LSTM layer. The classification model used for this purpose is RoBERTa supplemented with an LSTM layer. It learns contextualized word representations from a vast number of unlabelled text datasets. In the NLP evaluations, RoBERTa performs well because of its complex structure and excellent nonlinear representation learning capability. LSTMs can effectively boost performance since they can memorize and discover the pattern of crucial information. For classification purposes, we include a 128-node feed-forward linear layer. Inputs are normalized using a batch normalization layer. Following this, a dropout layer with a rate of 0.6 is implemented to mitigate the potential for overfitting. The input reviews are labeled as either fake or real thanks to the inclusion of the two feed-forward layers, each having an output size of the output of one cell and prioritized by selecting a threshold of 0.9 above the other. Fig. 3 depicts the overall architecture of the proposed model.

4. Performance evaluation

This section outlines the detailed analysis of the performed experiments, including the description of the datasets, the evaluation metrics, and the proposed model's performance. It also provides a performance comparison of the proposed model with respect to a number of state-of-the-art models.

4.1. Experimental setup

Training is performed for ten iterations across all datasets with a

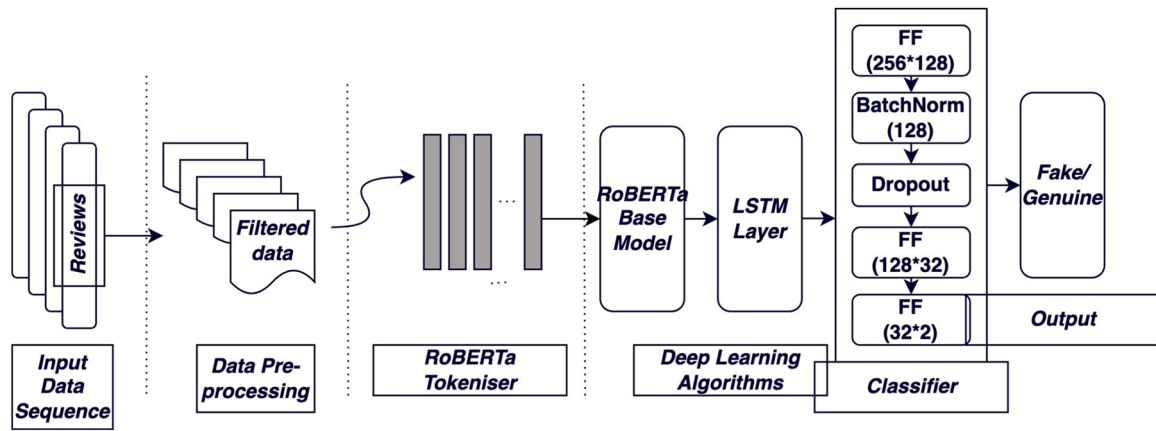


Fig. 3. The proposed model- RoBERTa with LSTM.

mini-batch size of 32. We use the early stop method based on validation loss to avoid the overfitting of the model as the $\Delta = 0$ (Prechelt, 1998). AdamW is employed to optimize the effectiveness of the proposed model (Loshchilov & Hutter, 2017). We adopt binary cross-entropy to calculate the loss finally (Rosasco, De Vito, Caponnetto, Piana, & Verri, 2004).

We program our model using Python language and the Google CoLab interface (Halyal, 2019). In addition to this, we utilize the newly developed transformer library (Wolf, et al., 2020). Several iterations of the tests are conducted, each with a unique set of parameters (Table 1).

4.2. Dataset description

Two publicly accessible benchmark datasets are utilized in this study. **Deception dataset** (Li, et al., 2014) and **OpSpam** (Ott, et al., 2011). Both data sets solely contain review text, with no corresponding metadata information. The **deception dataset** was constructed using crowdsourcing platforms. The dataset includes 3032 reviews and data about three different sectors (hotels, doctors, and restaurants). The **OpSpam dataset** was developed based on reviews related to the most prestigious hotels in Chicago. This dataset depicts a semi-real world. The dataset includes sixteen hundred text reviews for 20 hotels in Chicago, United States of America metropolitan region. Eight hundred reviews belong to fraudulent reviews, and the remaining eight hundred consist of authentic reviews. A label of "1" denotes that the reviews are false, while a "0" label specifies that the reviews are authentic. The testimonies come from a large number of distinct sources. The Amazon Mechanical Turk (AMT) platform is used to create the bogus reviews. In contrast, the remaining reviews are obtained from various online review sites such as Yelp, TripAdvisor, and Expedia. In the experiments, we use 80 % of the Deception and OpSpam datasets for training and 20 % of each dataset for testing. Table 2 presents the statistical data for both datasets.

4.3. Evaluation metrics

In the related work of detecting fake reviews, recall, accuracy, and the F-measure are the most popular performance metrics for evaluating the detection performance of various models are given that the number of reviews for both groups is equal. We use these metrics to evaluate the performance of the proposed model.

- **Accuracy:** This metric comprehensively estimates the number of cases that have been accurately identified, which can be determined as:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

where TP represents the "true positive", TN indicates the "true negative", FP is the "false positive", and FN is the "false negative".

- **Precision:** This metric describes the proportion of successfully predicted reviews to the total number of reviews for a given class which can be calculated as follows:

$$Precision(P) = \frac{\text{number of correct predictions of each class}}{\text{total number of predictions of each class}} = \frac{TP}{TP + FP} \quad (1)$$

- **Recall:** This metric shows the proportion of relevant reviews achieved from the total number of reviews and is calculated as follows:

$$Recall(R) = \frac{\text{number of correct predictions}}{\text{total number of predictions}} = \frac{TP}{TP + FN} \quad (2)$$

- **F1 score:** This metric shows the average of precision and recall and can be calculated as follows:

$$F - \text{measure} = \frac{2 \times \text{recall} \times \text{precision}}{\text{recall} + \text{precision}} \quad (3)$$

4.4. Results and discussion

4.4.1. Results for the op-spam dataset and deception dataset

Table 3 shows that the accuracy, precision, recall, and F1-score for the proposed model for the Op-Spam dataset are 96.03 %, 91.48 %, 99.31 %, and 95.36 %, respectively. The results of the experiments whereas for the deception dataset are 93.15 %, 85.39 %, 97.40 %, 91.20 %. The results of the experiments reveals that the proposed model performs well on OpSpam and Deception datasets. This is due to the model's capability of effectively capturing the complete contextualized depiction of each review, as well as its most pertinent characteristics, by employing a variety of models and perspectives.

4.4.2. Comparison with state-of-the-art models

To determine the efficacy of the proposed model, we compare its performance to a set of reference approaches, including SVM (Ott, et al., 2011), SVM (Cagnina & Rosso, 2015), SVM (Feng, et al., 2012), SAGE (Li, et al., 2014), RCNN (Lai, Xu, Liu, & Zhao, 2015), GRNN-CNN (Y. Ren & D. Ji, 2017), DRI-RCNN (Zhang, et al., 2018), LDA with TextCNN (Cao, Ji, Chiu, He, & Sun, 2020), MFNN (Jiang, Zhang, & Jin, 2020),

Table 1
Summary of existing works.

References	Model	Cons	Pros
(Shushkevich & Cardiff, 2021)	BiLSTM	By combining the assets of multiple models, the ensemble method mitigates the effect of individual flaws and can increase the model's overall precision.	The computational intensity and resource requirements of an ensemble may surpass those of a single model.
(Bandyopadhyay & DUTTA, 2020)	KNN	The KNN algorithm is straightforward and intuitive, making it simple to comprehend and implement.	The KNN algorithm exhibits laziness in learning, resulting in a notably lengthy prediction time in comparison to alternative algorithms.
(Felber, 2021)	SVM	Support Vector Machines (SVMs) have demonstrated their ability to accurately represent intricate decision boundaries and flourish when applied to high-dimensional datasets.	In particular, SVMs may incur significant computational costs when applied to vast datasets.
(Koloski, Perdih, Robnik-Sikonja, Pollak, & Skrlj, 2022)	BERT	Words with meanings that vary depending on context, idiomatic expressions, and long-range dependencies are all aspects of language structures that BERT is capable of processing.	Due to the substantial computational resources needed for both training and inference, BERT is unsuitable for deployment on systems with limited resources.
(Shifath, Khan, & Islam, 2021)	Foundation ensemble model	By combining the merits of numerous pre-trained transformer models, ensemble models of transformers can outperform them on subsequent tasks.	Large quantities of memory and computational expense are characteristics of ensemble models that make them challenging to implement on devices with limited resources.
(Duma, et al., 2024)	Hybrid approach	Integrating multi-data features (reviewer- and review-centric).	Large quantities of memory and computational expense are required.
(Thuy, et al., 2024)	DenyBERT model	Combining BERT and Knowledge Distillation techniques.	The proposed model focused only on English-language.
(Cheng, Wu, Chao, & Wang, 2024)	graph neural network	By combining different review text and reviewer features.	Large quantities of memory and computational expense are required.

SOM-CNN (Neisari, Rueda, & Saad, 2021), BERT (Thuy, et al., 2024) . The various reference models are described as follows:

- **SVM:** It is a classification model that combines bigram and LIWC characteristics using SVM.
- **SVM:** It is a classification model comprises of four grams and LIWC characteristics using SVM.

Table 2
Deception and OpSpam statistical information.

Datasets	Domain	Type of Reviews	Number of reviews	Number of words	Number of sentences
Deception (Li, et al., 2014)	Hotel reviews	legitimate	1080	17,328	9258
		Fake	1080	16,635	8463
	Doctor reviews	legitimate	200	5098	1151
		Fake	356	5128	2369
OpSpam (Ott, et al., 2011)	Restaurant reviews	legitimate	201	5126	1892
		Fake	201	5136	1827
	Hotels reviews	legitimate	800	14,812	7963
		Fake	800	14,427	7192

- **SVM:** It is a classification model for utilizing unigram features with SVM.
- **SAGE:** It is a classification model combines topic modeling with a generalized additive model.
- **RCNN:** It is a classification model that combines of recurrent and convolutional neural networks.
- **GRNN-CNN:** It is a hybrid classification model for detecting false reviews that combines a GRU and a CNN.
- **DRI-RCNN:** It is a classification model that is built on recurrent convolutional deep neural networks (DRI-RCNN) for identifying false reviews based on analyzing their context.
- **BERT-Base Case:** It is a classification model that is based on pre-trained a deep bidirectional text representation. This model can handle unlabelled input by concentrating on the right and left context at all levels.
- **LDA with TextCNN:** In this model, coarse-grained features are extracted with LDA and a backpropagation (BP) neural network, while fine-grained features are extracted with the help of neural networks. Based on the two fine-grained characteristics, it trains a support vector machine classifier to identify phony testimonials. This model consists of one fully linked layer and four progressively smaller filter layers (sizes 2, 3, 4, and 5).
- **MFNN:** This model is a multivariate feature-based multilayer perceptron (MLP) that learns both local and global information to identify false reviews. This model employs three-, four-, and five-width multi-kernel CNNs.
- **SOM-CNN:** This model uses self-organizing maps (SOM) to visualise reviews to spot false ones. This model consists of a 32-filter 3×3 convolutional layer and a 2×2 Max-pooling layer.

Table 4, Figs. 4, and 5 show that the proposed model outperforms the state-of-the-art approaches on deception and OpSpam datasets. Compared to conventional ML and deep learning models, the approach attained a high level of accuracy with a limited dataset. For instance, the proposed model obtains more than 90 % accuracy with limited training data, but deep learning could not reach this level. Hence, both classic and deep learning models require huge training datasets. Nevertheless, large-scale data collection is not always feasible. Hence, the transformer model is a pragmatic solution for small datasets.

5. Model explanation

Using ML or deep learning-based models can enhance the decision-making process for customers and businesses. However, determining confidence in individual predictions of these models is a challenging problem. For example, ML models cannot be trusted in sensitive research areas like medicine, defence, and finance due to the level of

Table 3
Classification reports for OpSpam and Deception datasets.

Model	Datasets							
	OpSpam				Deception			
	Acc	P	R	F1-score	Acc	P	R	F1-score
Proposed model	96.03 %	91.48 %	99.31 %	95.36 %	93.15 %	85.39 %	97.40 %	91.20 %

Table 4
Comparison of the Proposed model with the state-of-the-art methods.

Machine Learning Algorithms	Deception dataset		Op-Spam dataset	
	Accuracy	F1-score	Accuracy	F1-score
SVM (bigram and LIWC features)	79.33 %	82.83 %	82.89 %	82.09 %
SVM (unigram feature)	83.33 %	79.53 %	86.09 %	83.12 %
SVM (LIWC and four grams features)(	81.67 %	81.03 %	84.34 %	83.21 %
SAGA	81.82 %	79.38 %	83.10 %	82.23 %
RCNN	82.16 %	82.00 %	83.21 %	81.23 %
GRNN-CNN	83.34 %	82.86 %	84.15 %	84.17 %
DRI-RCNN	85.24 %	83.56 %	87.24 %	85.36 %
BERT Base Case	86.20 %	85.50 %	90.31 %	89.56 %
LDA with TextCNN	86.36 %	85.90 %	90.39 %	90.90 %
MFNN	85.20 %	84.70 %	88.72 %	89.10 %
SOM-CNN	85.90 %	85.01 %	89.23 %	90.01 %
RoBERTa +LSTM	93.15 %	91.10 %	96.03 %	95.36 %

complexity. Consequently, there is a need to explain ML models to make them more reliable and responsible. Thus, both customers and businesses can have more visibility into how machine and deep learning-based fake review detection models compute their final output and they therefore can trust the models and their decisions. Therefore, in this work, we provide an explainable fake review detection model. We use the post-hoc explainable technique to explain the obtained results by our models and why it achieves the best performance in detecting fake reviews. Specifically, we implement this technique on example_1 and example_2. LIME also provides explanations by learning an interpretable model locally around our prediction and assigning specific weight values to each feature. In this paper, we analyze the prediction results of the interpreter to identify key aspects of fraudulent reviews in the most

widely used dataset. Fig. 6 shows how common keywords like "luxury", "hotel," "Chicago," and "room," are used as features by neural network models to determine if reviews are genuine. When the model detects a fraudulent review, it gives that review a larger weight term as shown in Fig. 6. On the other hand, as demonstrated in Fig. 7, if the reviews are genuine, the model can correctly classify them based on these terms. In Fig. 7, the top 10 most relevant terms from the reviews are shown by bars of increasing size. These terms are also indicated in the original document with orange color for false and blue for genuine as shown in Figs. 6 and 7.

6. Complexity analysis and limitations

Detecting fake reviews is also challenging for machines since they must be able to differentiate between "genuine reviews" and "fake reviews". However, we employ a range of characteristics (e.g., review content and emotions) to accurately detect fake reviews. Thus, during the training, the greater the number of incorporated characteristics into the models, the better it detects fraudulent reviews. In this paper, we develop an approach focusing on deriving textual characteristics from review text to analyze the distinctions between fake and genuine review material. The experimental results revealed that contextual characteristics are crucial for distinguishing between bogus and genuine reviews. The results indicate that LSTM and pre-trained models considerably enhanced the detection of brief textual false reviews.

We note that the success of pre-trained language models relies heavily on two factors: (1) the scope and breadth of the data used in the training process and (2) the design of the model itself. RoBERTa's training included BookCorpus, Wikipedia, and CC-News to expand its coverage. Though, in the hybrid method, using RoBERTa with LSTM can enhance detection performance. This helps to manage the fake review features and context variance. The proposed hybrid model (RoBERTa with LSTM) is the best compared to the state-of-the-art methods and

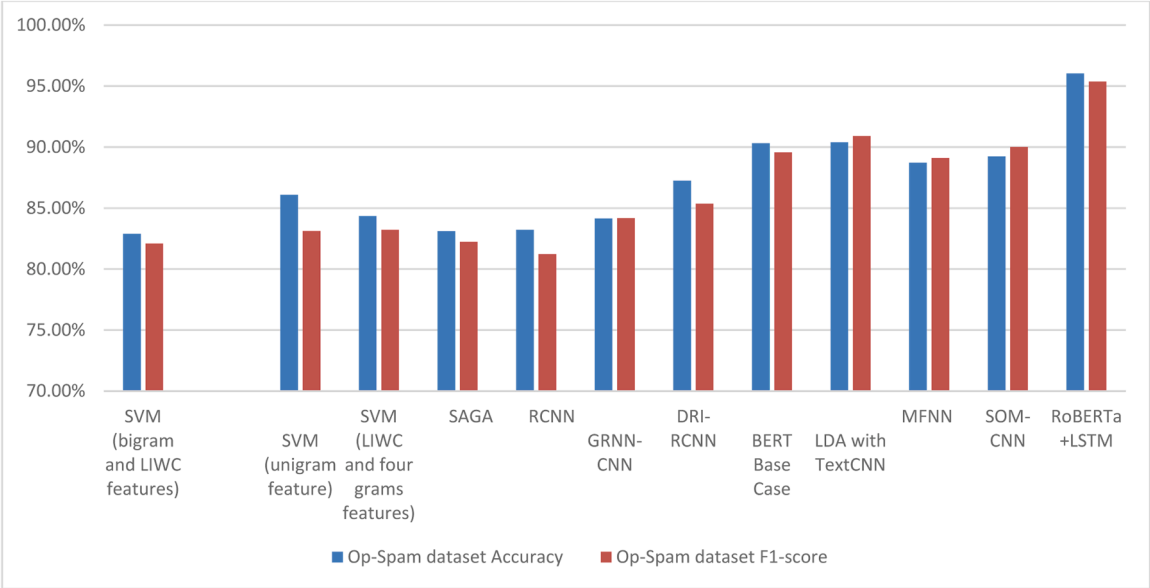


Fig. 4. Performance Comparison Under the Op-Spam dataset.

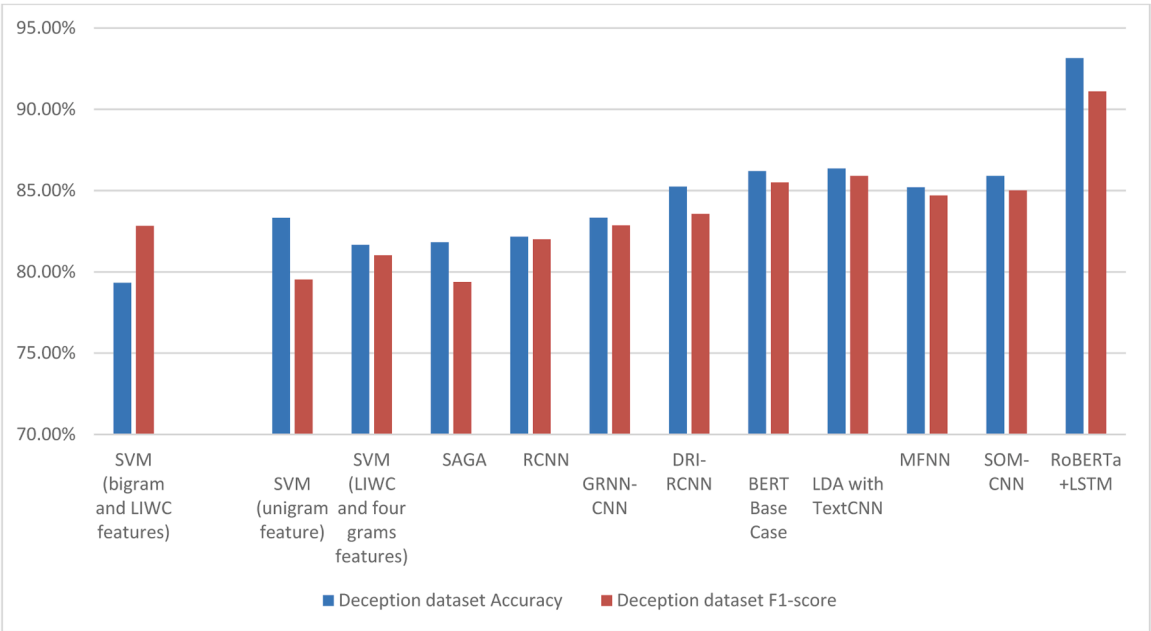


Fig. 5. Performance Comparison Under the Deception dataset.

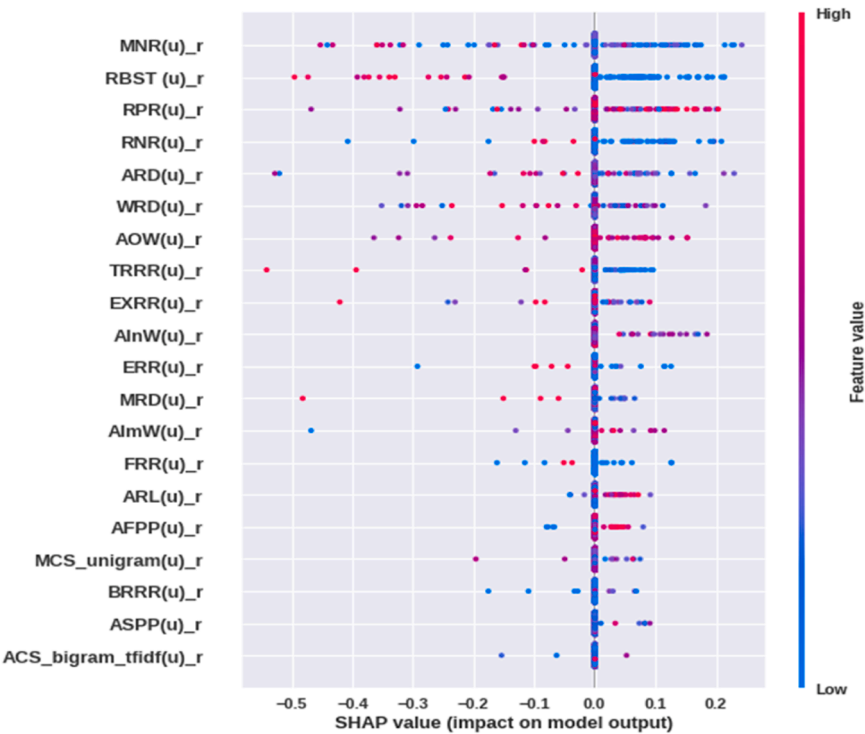


Fig. 6. Shape explanation on the Op-spam dataset.

clearly shows that our model outperforms the other proposed models. The findings reveal that the utilisation of LSTM and pre-trained models can significantly improve the ability to identify brief textual false evaluations.

Two factors significantly influence the effectiveness of pre-trained language models: (1) the extent and depth of the data utilised during training, and (2) the model's architecture. To broaden its scope, RoBERTa training incorporates BookCorpus, Wikipedia, and CC-News. With the hybrid approach, however, RoBERTa and LSTM can improve detection performance. This aids in the management of context

variation and false review features. When compared to the most recent methodologies, the performance of the proposed model unequivocally demonstrates that it surpasses the other models that have been proposed.

6.1. Complexity

Analysis the proposed model has an extremely low computational complexity due to the fact that it compares only pertinent terms extracted from the text, namely aspects and sentiments, with the

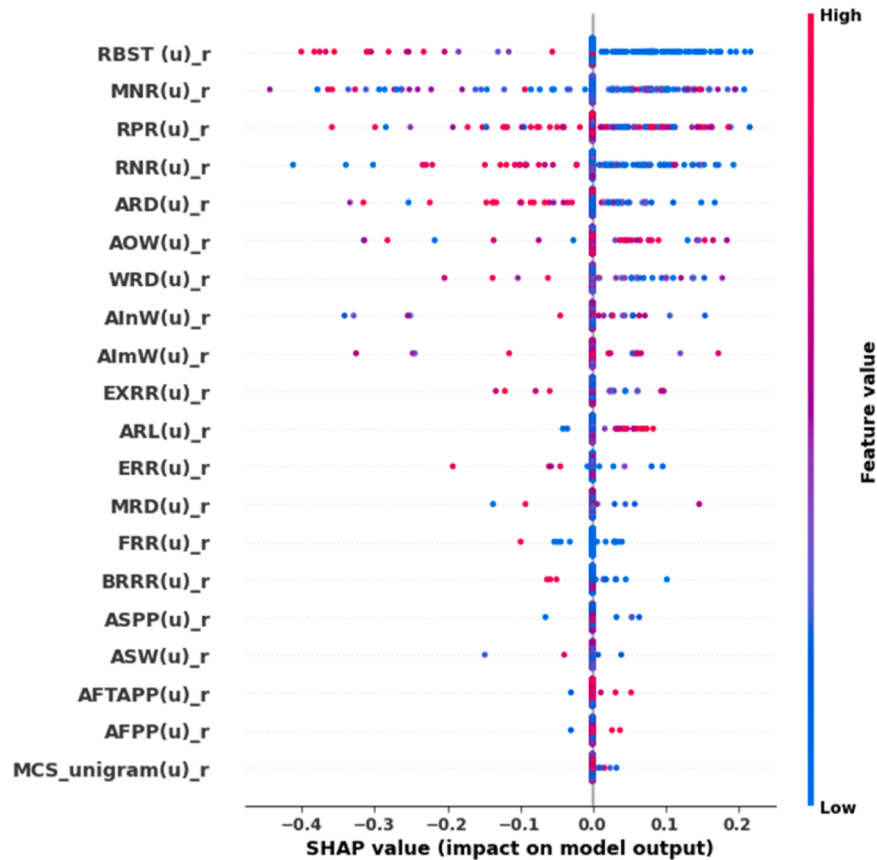


Fig. 7. Shape explanation on the deception dataset.

associated aspects and sentiments of other reviews. In our methodology, the computational complexity of reviews is $O(\log_2(n))$. Furthermore, the computational complexity of examining other reviews is $O(\log_2(n) \cdot \log_2(m))$. This procedure is to be iterated for k reviews, resulting in a total computational complexity of $O(k \cdot \log_2(n) \cdot \log_2(m))$. It is apparent from the computational complexity analysis that the proposed model significantly reduces the amount of time required to analyze a large number of reviews.

6.2. Limitation of this study

The findings presented in this research offer significant contributions to the understanding of the efficacy of different downstream neural network methodologies when it comes to detecting fake reviews utilizing transformer-based models. However, certain prospective limitations may be resolved in future research. For instance, the investigation failed to examine the effects of various optimization techniques and hyperparameters on the efficacy of the model. Furthermore, it is worth noting that the research employed a comparatively limited fake reviews dataset, which may restrict the applicability of the results. The impact of larger datasets and the applicability of the proposed models to related tasks could be the subject of additional study. Furthermore, an assessment of the models' interpretability may yield significant insights into the mechanisms that underlie them, thereby enhancing transparency and credibility. In addition, this study concludes that RoBERTa with LSTM performs better than alternative models. However, whether this result is attributable exclusively to the pre-training task or to other factors as well remains unknown. Hence, additional investigation is required in order to ascertain the precise elements that contribute to the exceptional performance of the proposed model. Further research may be warranted to examine the effects of various pre-training tasks and datasets in order to ascertain the proposed models' robustness.

7. Conclusion

This paper proposed a novel semantic- and linguistic-aware model for fake review detection that integrates a RoBERTa model with an LSTM layer. This approach leverages transformer architecture to accurately identify patterns in fake reviews, enhancing both fake and authentic behavior profiling. The proposed model significantly outperformed state-of-the-art models on semi-real benchmark datasets. This paper provides substantial contributions to deep learning, natural language processing, and online commerce security. The reported findings shed light on textual data processing, feature extraction techniques, and the detection of fraudulent online reviews. Reliable methods for detecting fake reviews strengthen the credibility of e-commerce, making this a critical objective for online commerce platforms. Moreover, the proposed work serves as a benchmark for future model developers using machine learning and deep learning techniques to identify fraudulent reviews. Despite the comprehensive experiments presented, further improvement is possible. Future research will evaluate the proposed method on cross-domain datasets and explore applying SHAP and deep learning insights to develop an interpretable lexicon-based classifier. We also plan to investigate combining multiple classifiers and feature extraction techniques to improve accuracy on larger datasets. Additionally, we will apply advanced textual analyses, such as named-entity recognition, to extract more valuable information and examine the effects of adjusting RoBERTa's hyperparameters and subsequent layers.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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